

Equal Everywhere Matrix

Input file: standard input
 Output file: standard output
 Time limit: 1 second
 Memory limit: 256 megabytes

Given an integer n and an $n \times n$ matrix. Your task is to check whether the matrix is a "**Equal Everywhere**" matrix or **not**.

1	2	3
4	5	6
7	8	9

1	2	3
4	5	6
7	8	9

Main Diagonal 1

1	2	3
4	5	6
7	8	9

Main Diagonal 2

*

Input

$n + 1$ lines of input:

The first line contains an integer n ($2 \leq n \leq 10$) representing the matrix dimensions.

the next n lines contains matrix values $matrix_{11}, matrix_{12}, \dots, matrix_{nn}$ ($0 \leq matrix_{ij} \leq 1000$).

Output

Print "**YES**" if the matrix is **Equal Everywhere**, otherwise, print "**NO**".

Examples

standard input	standard output
3 7 4 3 5 1 5 6 13 10	NO
3 10 10 10 10 10 10 10 10 10	YES

*An Equal Everywhere matrix is a matrix containing elements such that the sum of every row, every column, and both diagonals is the same.